Team : Go Paloks

Livestock Management System

CSE 4408: System Analysis and Design Lab

Presentation 10

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Project Context

Core System

- Integrated Farm Management System
- Inventory, Operations, Finance, Sales Tracking
- Livestock Health Monitoring System

Main Goal

- Improve Efficiency
- Reduce Wastage
- Prompt Alert and Decision Making
- Real Time Monitoring

Today's Agenda

- Candidate Primitive Processes
- Process Specification Form
- Structured English
- Decision Table & Decision Tree
- Ambiguity Handling

Why Process Specification

DFD shows the "WHAT"

- External entities
- Data flows
- Data stores
- High-level process movement

Process Spec shows the "HOW"

- Internal business rules
- Step-by-step processing logic
- Decision outcomes
- Code-ready documentation

Candidate Primitive Processes

DFD No.	Primitive Process	Reason for Consideration
2.1	**Capture Daily Feed Log**	Start of daily feed entry flow
2.2	**Validate Batch & Quantity**	Multiple validation conditions
2.3	**Generate Task Schedule**	Task generation from operation status
2.4	**Update Daily Operation Log**	Sequence with repeated feed-line updates
5	**Process Sales & Availability**	Availability decision and order outcome

Selected for formal specification:

- **2.4 : Structured English**
- **2.2 : Decision Table**
- **5 : Decision Tree**

Process Specification Form : Update Log (I)

Process: Update Daily Operation Log

Process No.
2.4

Input

Validated Feed Log
Task Completion Status

Output

Updated Operation Log
Operation Summary

Data Stores

D2 Operation Logs
D3 Inventory Records

Process Specification Form : Update Log (II)

Process: Update Daily Operation Log

Trigger

Farm worker feed log is accepted

Precondition

Feed batch and livestock group are valid

Postcondition

Daily operation record is saved

Business Rule

Every feed usage line must be recorded under:

- correct date
- worker
- farm section
- livestock group.

Technique Used

Structured English
Best for sequential processing with a small loop.

Structured English : Update Log (I)

Why suitable?

- Mostly sequential
- Contains one simple loop
- Easy to translate into code

Logic Summary

Valid feed entries are:

- appended into daily logs
- summarized for reports.

Structured English : Update Log (II)

```
UPDATE Daily Operation Log

RECEIVE Validated Feed Log

READ Current Operation Log for Feeding Date

IF Operation Log does not exist
    THEN create new Daily Operation Record
ENDIF

FOR EACH Feed Usage Line in Daily Feed Log Record
    ADD feed usage entry to Operation Log
    LINK entry with Worker, Section, and Livestock Group
ENDFOR

CALCULATE total feed consumed for the day

STORE updated Operation Log

GENERATE Operation Summary

END UPDATE Daily Operation Log
```

Process Specification Form : Validate Batch & Quantity (I)

Process: Validate Batch & Quantity

Process No.
2.2

Input

- Daily Feed Logs
- Required Feed Data

Output

- Validated Feed Log
- Update Stock
- Exception Notice

Data Stores

- D1 Farm Structure
- D3 Inventory Records

Process Specification Form : Validate Batch & Quantity (II)

Process: Validate Batch & Quantity

Trigger

Worker submits feed usage data

Precondition

Required fields are present in the feed log

Postcondition

Feed log is either accepted or returned

Business Rule

The system accepts feed usage only when

- livestock group
- feed batch
- quantity
- stock level

are all valid.

Technique Used

Decision Table

- Best for several interacting conditions and clear rule patterns.

Decision Table : Validate Batch & Quantity

Conditions / Actions	R1	R2	R3	R4	R5
Livestock group valid?	N	Y	Y	Y	Y
Feed batch exists?	–	N	Y	Y	Y
Quantity greater than zero?	–	–	N	Y	Y
Stock sufficient?	–	–	–	N	Y
Reject: invalid livestock group	X				
Reject: unknown feed batch		X			
Return for quantity correction			X		
Hold log + create stock exception				X	
Accept validated feed log					X
Update inventory record					X

Simplification: "–" means do not care. Invalid prerequisite rules stop early, so redundant combinations are merged.

Process Specification Form : Process Sales & Availability (I)

Process: Process Sales & Availability

Process No.
5

Input

- Product Orders
- Inventory Data

Output

- Availability
- Order Confirmation / Rejection

Data Stores

- D5 Sales Orders
- D3 Inventory Records

Process Specification Form : Process Sales & Availability (II)

Process: Process Sales & Availability

Trigger

Buyer submits product order

Precondition

Buyer details and product items are available

Postcondition

Order is confirmed, partial, or rejected

Business Rule

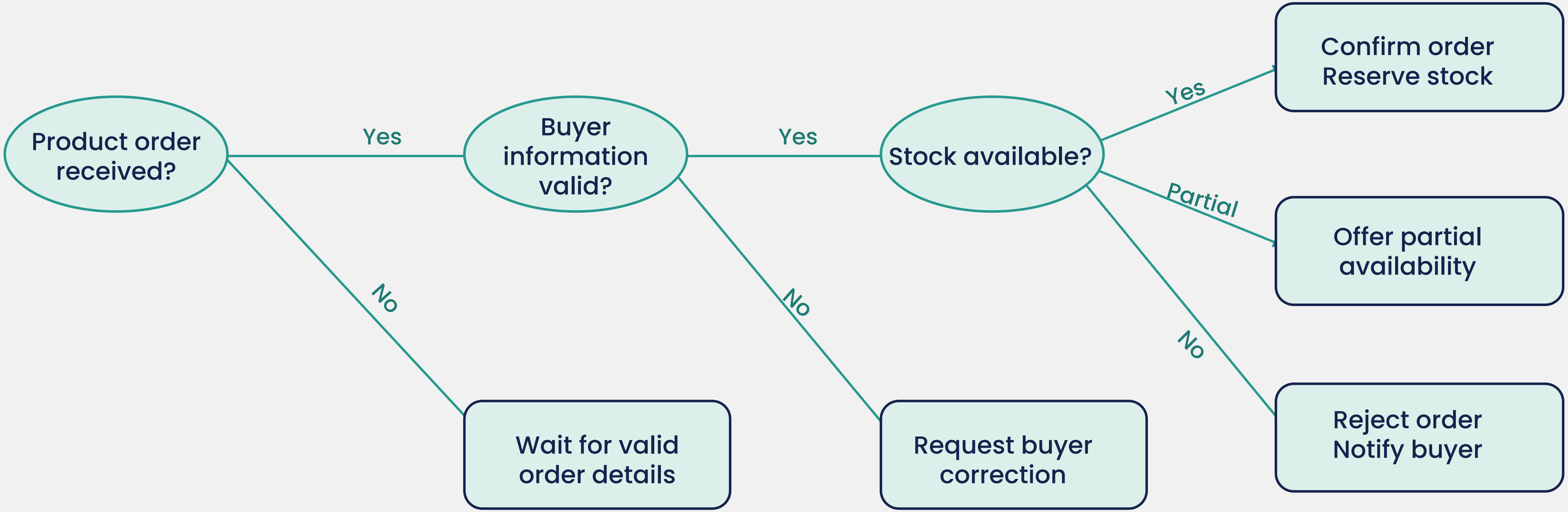
- Availability must be checked before confirming sales.
- The system may confirm, offer partial fulfillment, or reject.

Technique Used

Decision Tree

- Best for ordered decisions with different branch lengths.

Decision Tree : Process Sales & Availability



Technique Selection Summary

1

Structured English

2.4-Update Daily Operation Log

Best for ordered steps and small repeated feed-line updates.

2

Decision Table

2.2-Validate Batch & Quantity

Best for several conditions that create clear rule patterns.

3

Decision Tree

5-Process Sales & Availability

Best for ordered branches where outcomes are visually traceable.

Ambiguities and Issue Handling

Issue Found

- Exact low-stock threshold not finalized
- Partial order policy is unclear
- Who approves rejected/held feed logs?

Current Assumption

- Low stock alert triggers when requested feed exceeds available stock
- Partial fulfillment is allowed only when buyer accepts reduced quantity
- Farm Manager reviews held or rejected transactions

Next Action

- Confirm threshold with farm manager
- Add buyer confirmation rule in sales form
- Define approval authority and audit trail

Thank You

We are ready for your valuable feedback!